

HTX xData | Training Report

Task 4 | Final experimental review | 13 September 2026

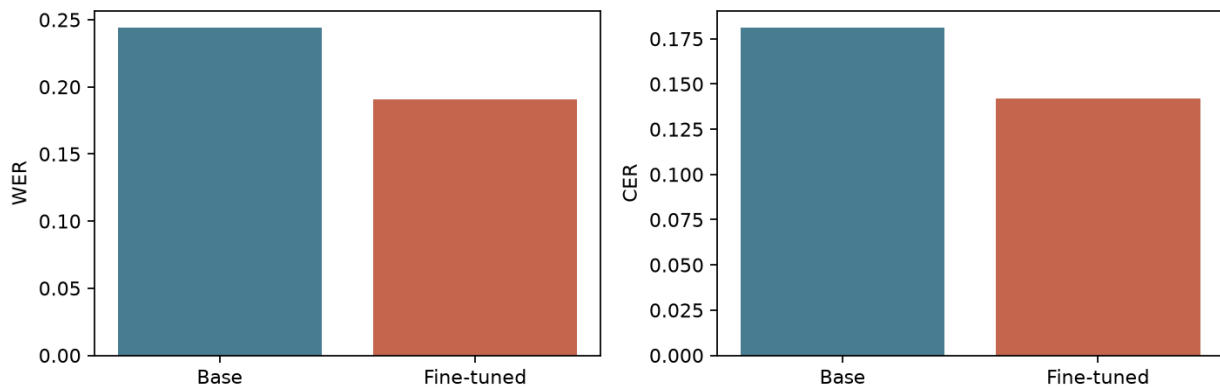
1. Full Daily Ketchup results

E3 is the selected submission model, chosen using non-TDK validation. Its complete evaluation preserves all 218,011 rows across 367 videos. The public E3 release, model, prediction CSV and artifact manifest match. E1 is retained as a historical release and comparison, not the default downloadable model.

Metric	Base	E1 historical	E3 selected
Corpus WER	24.4272%	21.3120%	19.0689%
Corpus CER	18.1281%	15.8369%	14.2119%
Paired coverage	218,011	218,011	218,011

E3 reduces WER by 5.3583 percentage points versus base, a 21.9360% relative reduction. These are pooled corpus edit rates, not averages of per-row or shard WER. No rows were removed to improve the score.

The 2,000-draw video-cluster bootstrap gives a 95% interval [-5.4961, -5.2234] percentage points for E3-minus-base WER. Recurring speakers and related episodes may violate video independence; this interval does not establish performance on all Singapore English.



Design and Controlled Experiments

2. Data and preprocessing

All Daily Ketchup clips were excluded from training and validation. E1 and E3 share exactly hashed manifests: 10,972 training clips / 10.046 hours / 852 source videos; 2,133 validation clips / 2.023 hours / 93 disjoint videos. Selection was deterministic and channel-balanced. Video-disjoint does not imply speaker-disjoint.

Metadata boundaries and transcript validity were checked; decoded audio was validated and converted to mono 16 kHz. The pretrained 8,192-token SentencePiece tokenizer and 128-bin log-mel frontend were retained. No denoising, VAD trimming, UTMOS/SIGMOS filtering, or tokenizer replacement was performed. Those are candidate ablations, not claimed completed steps.

3. Fine-tuning and selection

Full-parameter PyTorch/NeMo adaptation: AdamW, learning rate 1e-5, betas (0.9, 0.98), weight decay 0.001; BF16; batch 8 with accumulation 4; clipping 1.0; seed 2026. Validation is checked every 50 optimizer updates. The same validation set selects checkpoints; TDK is not used for tuning.

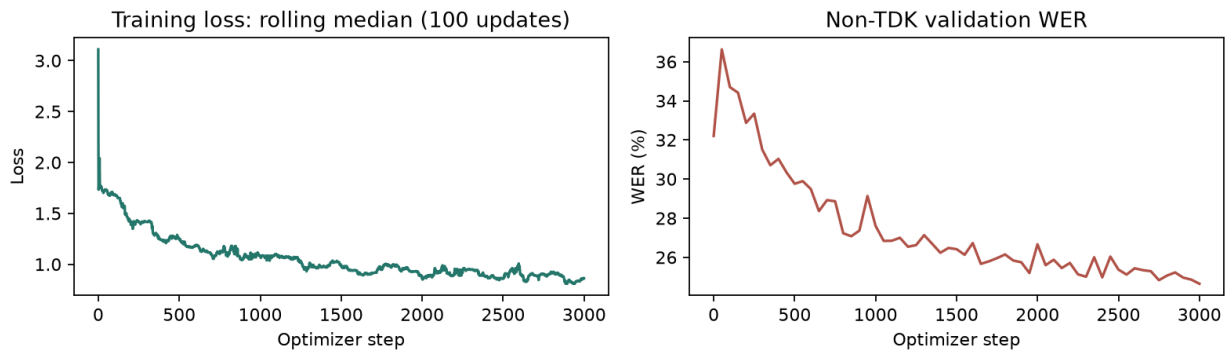
Experiment	Training / budget	Native validation WER
E1	10.046 h; 1,200 updates	26.5402%
E2	25.046 h; 1,200 updates	26.6967%
E3	E1 data; 3,000 updates	24.6470%
E4	Resume E3; 380 additional updates	Excluded: inconsistent selection metric

E2 does not prove more data is harmful: equal optimizer updates imply less exposure per example. E3 supports extending adaptation on the fixed data, but a single seed does not establish optimal hyperparameters. E3 reached its best checkpoint at the step cap after 160.79 fit minutes; convergence is not established.

Audio provenance

Original clip URL outages required recovery from original YCSEP WAV timestamp ranges and MP3 encoding. The baseline used real HTTP /asr calls in Azure. E3 reused the same hash-verified cached MP3s. They are not byte-identical to the original audio-column MP3s. Merge checks bind all rows to the baseline hashes, correct shard indices and the downloaded E3 model SHA-256.

Learning Curves and Error Analysis



The loss plot shows a 100-update rolling median to reduce batch noise. Validation improves over the extended horizon but fluctuates; falling training loss alone is not evidence of better recognition. Native NeMo validation WER and normalized TDK WER are different measurement contexts and should not be directly compared.

4. E4 is not valid evidence of convergence

The continuation restored optimizer state and also inherited the previous Timer state, stopping after 20.82 new fit minutes. It recorded 14.93% at the unchanged resume step although the starting full validation was 24.65%. Later checks were around 24.61-25.28%. The selected artifact and 14.93% result are excluded pending validation-loop investigation. This is an implementation limitation, not a demonstrated model gain or plateau.

5. Where errors remain

Clip duration	Rows	Base WER	E3 WER
<2s	117,758	49.8087%	37.5281%
2-5s	62,338	24.2971%	19.2037%
5-10s	29,134	16.9546%	13.6281%
>=10s	8,781	13.6598%	10.8692%

Video outcomes: improved: 367. These are post-test descriptive slices, not checkpoint-selection criteria. Short fragments, overlap and annotation alignment are plausible causes of errors, not manually established labels.

Both models use Unicode NFKC and case-folded word normalization with internal apostrophes; CER excludes spaces. Case and punctuation quality are not measured. WER can exceed 100% for individual short references. Largest-change error examples are diagnostic rather than representative.

Improvement Plan and Verdict

6. Stop expansion; prioritize evidence

The completed work supports a useful adaptation gain, not convergence or a globally optimal model. Further broad GPU runs are not justified for this deadline. Finalize consistent model/data artifacts, executable notebooks and provenance first. Do not repeatedly optimize against the now-inspected TDK set.

Priority	Controlled next experiment
Alignment and coverage	Label a stratified non-TDK sample across channel, duration and base-error bands. Separate bad annotations from valid difficult Singlish; high WER alone is not a deletion rule.
Data exposure	Compare nested 10-hour/25-hour YCSEP sets at both matched updates and matched exposure. Keep validation fixed; use multiple seeds where affordable.
Learning rate and duration	Repair and regression-test resumed validation/timer state before another continuation. Use a bounded learning-rate comparison (5e-6, 1e-5, 2e-5) and validation early stopping; preserve the clean control.
Context and audio	Test adjacent same-speaker concatenation with alignment checks. Compare mild noise augmentation or denoising separately against unchanged audio, rather than adding all preprocessing at once.
External data	After access/license and duplication checks, test a bounded mixture with IMDA National Speech Corpus. Evaluate on new untouched recordings and a channel/speaker-disjoint set.

References and reproducible evidence

Dataset: YCSEP, Harvard Dataverse, DOI 10.7910/DVN/B7JRID. Base model: NVIDIA nvidia/parakeet-tdt-0.6b-v3 on Hugging Face. Proposed external data: IMDA National Speech Corpus (not used in the reported experiments).

Recorded evidence: asr-train/results/E1/, E1-TDK/, E3/ and E3-TDK/. E3-TDK contains paired comparison, corpus counts, video-level outcomes, error examples and strict evaluation provenance. E4 raw logs are retained in local working records; its suspect score is not a submission result.